HS 224 Game Theory and Economics

I am your course instructor, Debarshi Das. This course will introduce you to some preliminary ideas of Game Theory. Game Theory analyses situations of *strategic interaction* between multiple decision makers. Examples of such strategic interaction can be from the field of Economics. For example, how do companies decide the price of their product?

The examples could be from the field of Political Science. For example, in an election how do candidates decide which position they will take on an issue?

From Biology, for example, how animals deal with competitors when they fight over prey?

Auction Theory, for example, which bid is the best in an auction?

Game Theory has found application in the field of engineering as well. Although this course is titled 'Game Theory and Economics' we shall discuss the use of Game Theory in fields other than Economics.

This is an introductory course, therefore the topics which will be covered will give only a glimpse of the vast literature. We shall discuss **non-cooperative games with perfect information**. In this class of games the decision makers behave individually, not as a group.

The course has a **major emphasis on applications**. For this purpose plenty of examples and exercises will be covered. Roughly 40 lectures will be used to do justice to the topics. Extra classes will be arranged as per requirement and convenience.

The course **does not have any prerequisite**. However it is advised that students brush up their knowledge of differential calculus and probability theory.

Evaluation and weights: We shall aim to have two quizzes, one before the midsem and one before the endsem exam. If we have time for two quizzes, the weights on the tests would be: **10-40-10-40**. If we run short of time (which often happens!) we shall skip the second quiz. In that case the weights would be **10-40-50**. Remember the absolute marks in tests do not matter, what matters are the weights. The final grading will be relative.

Brief Outline of Topics:

<u>Strategic games: Theory</u>

Theory of rational choice, strategic games with perfect information: what are they, simple examples, Prisoner's Dilemma, Battle of Sexes, Matching Pennies, Stag Hunt, Coordination games. Nash equilibrium as a solution concept, finding Nash equilibrium, dominance, best response functions and Nash equilibrium, strong dominance and weak dominance of actions, symmetric games, symmetric equilibrium. Other solution concepts: dominance solvability, rationalizability.

Strategic games: Applications

Nash equilibrium in the Cournot model of oligopoly markets. Effect of increasing competition, technological improvement. Bertrand model oligopoly of oligopoly markets. Choices made by candidates in an election. Wars of attrition. Different sorts of auctions: second price sealed bid auction, first price sealed bid auction and their Nash equilibria. Lobbying as auction. Laws of compensation in an accident.

Mixed strategy games

Mixed strategy as a general concept of pure strategy. Properties of mixed strategy Nash equilibrium. Finding the mixed strategy Nash equilibrium in the simple 2*2 games and in more complex games.

Extensive games: theory and applications

Definition of extensive games or sequential games with perfect information. Nash equilibrium in such games and its conceptual difficulties. Idea of subgame perfect equilibrium, and finding it. Finite horizon games and use of backward induction. Applications: ultimatum game, Stackelberg model of duopoly markets.

Readings:

Mainly, I shall be following:

> M. J. Osborne, *An Introduction to Game Theory*, Oxford University Press, 2004 (the first three chapters of this book are freely available on the internet).

However, from time to time I shall depend on:

> *Games of Strategy* (by A. Dixit, S. Skeath, D. Reiley Jr., Viva Books, Third Edition, 2010)
>
> *Microeconomic Theory* (by A. Mas-Colell, M. D. Whinston, J. R. Green, OUP, 1995).